

Heart Disease Prediction Using Machine Learning

Abstract

This internship capstone project develops a machine-learning workflow to predict the presence of heart disease from clinical patient data. The work uses the combined UCI Heart Disease dataset supplied with the project, containing 920 raw records and 16 columns. After removing the identifier column and 2 duplicate rows, the modeling dataset contained 918 records. The original severity target, num, was converted into a binary target: 0 for no disease and 1 for disease present.

The pipeline handled missing values through median imputation for numeric variables and mode imputation for categorical variables, label-encoded categorical features, applied an 80/20 stratified train/test split with random_state=42, and standardized features using StandardScaler fitted on the training data only. Logistic Regression and a Decision Tree Classifier (max_depth=5) were trained and evaluated on the same held-out test set of 184 records.

Both models achieved accuracy of 0.8152. Logistic Regression was selected as the final model because it achieved the stronger ROC AUC of 0.9073, compared with 0.8684 for the Decision Tree, and higher recall of 0.8627. The project demonstrates a complete, reproducible baseline workflow while also identifying limitations related to substantial missingness in some clinical variables and the need for external validation before any real clinical use.

1. Introduction

Heart disease prediction is a binary classification task in which patient clinical measurements are used to estimate whether heart disease is present. In this project, classical supervised learning models are trained using structured records that include demographic information, chest-pain category, blood pressure, cholesterol, electrocardiographic results, exercise-related variables, and other clinical measurements.

The project focuses on building an understandable end-to-end workflow rather than using a complex model. It includes data inspection, cleaning, exploratory data analysis (EDA), preprocessing, training, evaluation, visualization, model comparison, and a sample prediction. The goal is to compare two commonly used baseline classifiers—Logistic Regression and Decision Tree—and select a model using held-out test-set performance.

2. Problem Statement

The task is to predict whether heart disease is present for a patient based on the available clinical feature columns. The source target column, num, represents severity levels from 0 to 4. The executed project converts this into a binary outcome: target = 0 when num = 0 and target = 1 when num > 0. The resulting model output is therefore a disease-presence prediction, not a severity-stage prediction or a clinical diagnosis.

3. Objectives

- Inspect the supplied Heart Disease UCI dataset for structure, data types, missing values, duplicate rows, and target distribution.
- Prepare a binary target for disease presence and create a usable modeling dataset without silently discarding most records.

- Apply documented preprocessing: duplicate removal, type conversion, missing-value imputation, categorical encoding, and feature scaling.
- Train and compare Logistic Regression and Decision Tree classifiers using a stratified held-out test set.
- Evaluate the models using accuracy, precision, recall, F1-score, confusion matrices, ROC curves, and ROC AUC.
- Select and save the model with the higher ROC AUC and demonstrate its output on one held-out test record.

4. Dataset Description

The project uses the supplied combined Heart Disease UCI dataset. It includes records from Cleveland, Hungary, VA Long Beach, and Switzerland. The raw file has 920 rows and 16 columns: an id column, 14 clinical feature columns, and the num target column.

Property	Observed value
Dataset	Heart Disease UCI dataset (combined multi-center version)
Raw dataset size	920 rows × 16 columns
Rows after duplicate removal	918
Duplicate rows removed	2 (after excluding id from duplicate check)
Model input features	14
Original target	num: 0 = no disease; 1–4 = increasing severity
Modeling target	target: 0 = no disease; 1 = disease present (num > 0)
Binary class counts after cleaning	No disease: 410; Disease: 508
Train/test split	734 training rows; 184 test rows

The 14 modeling features are age, sex, dataset, cp, trestbps, chol, fbs, restecg, thalch, exang, oldpeak, slope, ca, and thal. The id column is an identifier rather than a clinical predictor and was dropped before modeling.

Column	Missing values in raw dataset
trestbps	59
chol	30
fbs	90
restecg	2
thalch	55
exang	55
oldpeak	62
slope	309

Column	Missing values in raw dataset
ca	611
thal	486

The missingness is especially important for ca and thal, which have 611 and 486 missing values respectively. These values were retained through imputation rather than deleting affected rows, because listwise deletion would remove a large portion of the available data.

5. Data Preprocessing

The preprocessing choices below are taken directly from the executed Python script and notebook.

- Dropped id because it is a row identifier and not a predictive clinical feature.
- Removed 2 exact duplicate rows after id was excluded.
- Converted fbs and exang values stored as TRUE/FALSE into 1/0.
- Created target by binarizing num: num = 0 became 0; num > 0 became 1.
- Imputed numeric columns (age, trestbps, chol, thalch, oldpeak, ca) with their median values.
- Imputed categorical columns (sex, dataset, cp, fbs, restecg, exang, slope, thal) with their mode values.
- Label-encoded the categorical columns so that the classifiers could consume numeric inputs.
- Used train_test_split with test_size=0.2, stratify=y, and random_state=42.
- Applied StandardScaler, fitted only on the training data and then used to transform both training and test features.

After preprocessing, the script reports zero missing values. The scaling step was particularly relevant to Logistic Regression; the same scaled representation was also passed to the Decision Tree for a consistent comparison pipeline.

6. Exploratory Data Analysis

EDA was performed after cleaning and imputation. The project generated a class-distribution chart, six numeric feature distributions split by target class, and a correlation heatmap of numeric variables plus the binary target.

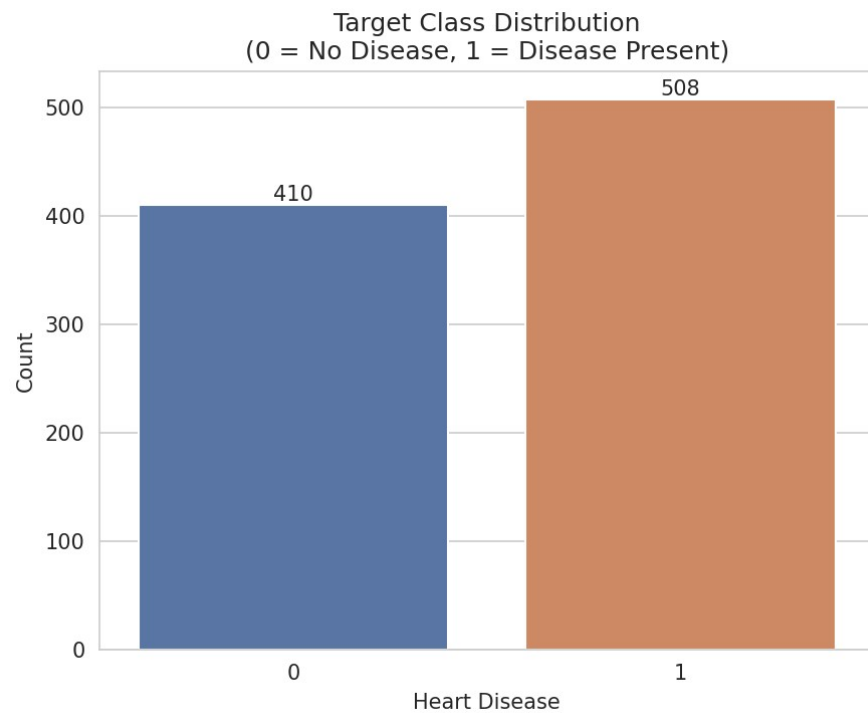


Figure 1. Binary target distribution after cleaning: 410 records with no disease and 508 records with disease present.

The target distribution is moderately balanced rather than extremely skewed. Stratification was nevertheless used during the train/test split so that the relative class balance was preserved in both subsets.

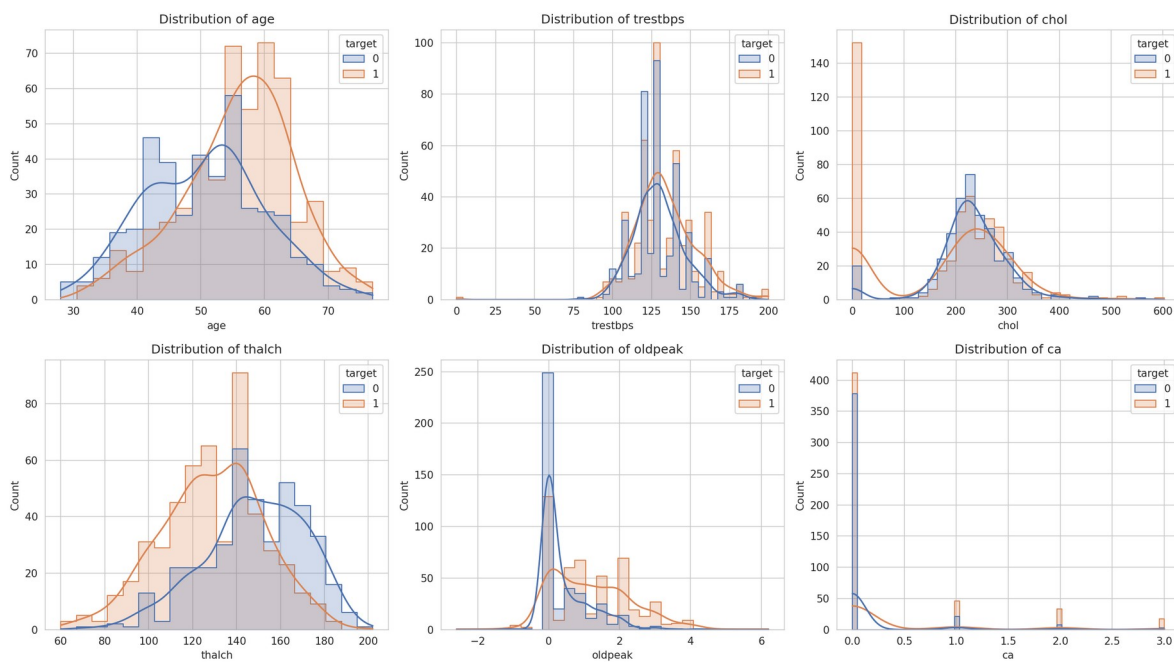


Figure 2. Distributions of the numeric features age, trestbps, chol, thalch, oldpeak, and ca, shown separately by target class.

These plots provide a visual check of how the numeric measurements are distributed across the two target classes. They are exploratory; the project does not claim causal relationships or use these graphs to remove features.

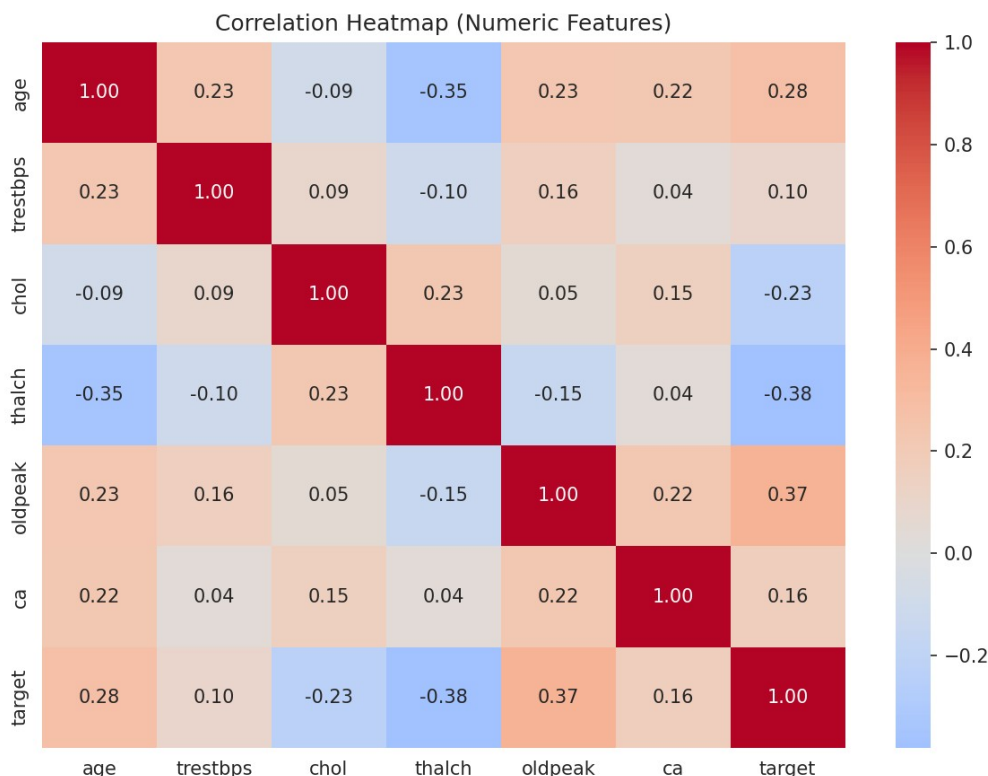


Figure 3. Correlation heatmap for numeric features and the binary target.

The heatmap summarizes pairwise linear correlations among the numeric variables and target. It was used as an exploratory view only; no feature-selection procedure was executed based on the correlation values.

7. Methodology / ML Workflow

The implemented workflow follows a standard supervised binary-classification process:

- Load the CSV dataset and inspect its shape, column types, missing values, duplicates, and original target distribution.
- Clean the data by removing id and duplicate rows, fixing Boolean-like columns, and creating the binary target.
- Impute missing numeric and categorical values using median and mode respectively.
- Generate EDA visualizations.
- Separate predictors (X) from the target (y), then label-encode categorical features.
- Create stratified 80/20 training and test partitions with random_state=42.
- Fit StandardScaler on X_train only and transform X_train and X_test.
- Fit Logistic Regression and Decision Tree models.
- Generate class predictions and positive-class probabilities on the held-out test set.
- Compute metrics and visualizations, compare models, choose the higher-AUC model, save the model bundle, and run one sample prediction.

8. Logistic Regression

Logistic Regression was trained with `random_state=42` and `max_iter=1000`. It received the standardized training features. The model produces both a binary class prediction and a probability for the disease-present class.

Metric	Logistic Regression result
Accuracy	0.8152
Precision	0.8148
Recall	0.8627
F1-score	0.8381
ROC AUC	0.9073

Its ROC AUC was the highest of the two evaluated models. Because the project explicitly selects the final model using ROC AUC, Logistic Regression was saved as the selected model.

9. Decision Tree

The Decision Tree Classifier was trained with `random_state=42` and `max_depth=5`. Limiting maximum depth was the implemented strategy to reduce overfitting. The model was evaluated on the same standardized held-out test data as Logistic Regression.

Metric	Decision Tree result
Accuracy	0.8152
Precision	0.8542
Recall	0.8039
F1-score	0.8283
ROC AUC	0.8684

The Decision Tree achieved the same accuracy as Logistic Regression and a higher precision, but it produced lower recall, lower F1-score, and lower ROC AUC. Therefore, it was not selected as the final model under the project's stated selection rule.

10. Evaluation Metrics

All metrics were calculated on the held-out test set of 184 records. Accuracy measures the overall proportion of correct predictions. Precision measures how often a disease prediction was correct. Recall measures the proportion of actual disease cases detected. F1-score combines precision and recall. The confusion matrix reports counts of correct and incorrect predictions. ROC curves plot true positive rate against false positive rate across thresholds, and ROC AUC summarizes the ranking performance over those thresholds.

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Logistic Regression	0.8152	0.8148	0.8627	0.8381	0.9073
Decision Tree	0.8152	0.8542	0.8039	0.8283	0.8684

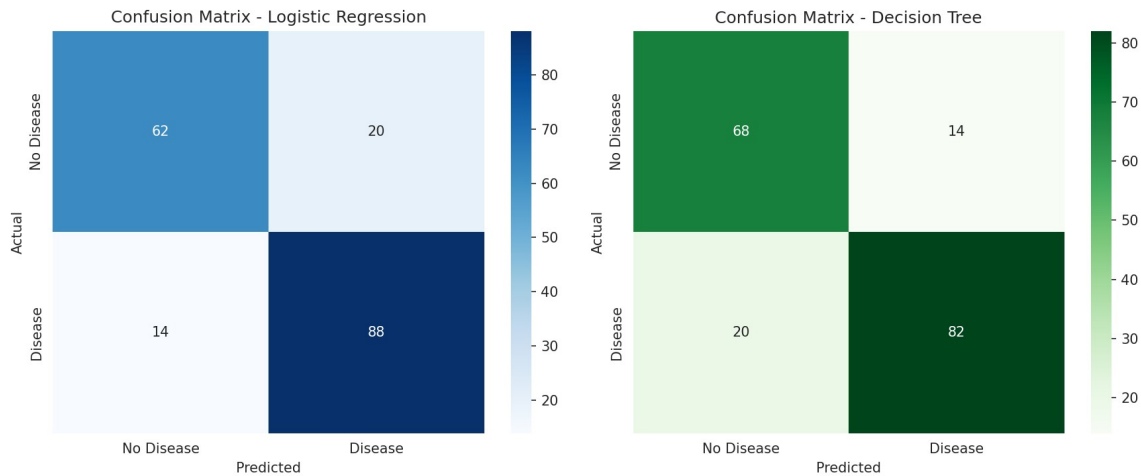


Figure 4. Confusion matrices for Logistic Regression and Decision Tree. Rows are actual classes; columns are predicted classes.

11. Results and Model Comparison

The two models tied on accuracy at 0.8152. The performance comparison therefore relies on the full metric set rather than accuracy alone. Logistic Regression had higher recall (0.8627 versus 0.8039), higher F1-score (0.8381 versus 0.8283), and higher ROC AUC (0.9073 versus 0.8684). The Decision Tree had higher precision (0.8542 versus 0.8148).



Figure 5. Side-by-side comparison of accuracy, precision, recall, F1-score, and ROC AUC.

Logistic Regression was selected because the implemented selection step chooses the model with the highest test ROC AUC, using F1-score only as a tiebreaker. Its stronger recall also means it missed fewer disease-present records in this held-out test result.

12. Confusion Matrix Analysis

Model	True negatives	False positives	False negatives	True positives
Logistic Regression	62	20	14	88
Decision Tree	68	14	20	82

For Logistic Regression, the confusion matrix was [[62, 20], [14, 88]]. It correctly identified 88 disease-present cases and 62 no-disease cases. It produced 20 false positives and 14 false negatives.

For the Decision Tree, the confusion matrix was [[68, 14], [20, 82]]. It produced fewer false positives than Logistic Regression (14 versus 20), which is consistent with its higher precision. However, it produced more false negatives (20 versus 14), which is consistent with its lower recall.

In a screening-oriented interpretation, false negatives are important because they represent disease-present cases predicted as no disease. The report does not assign clinical costs or propose a threshold change, but the observed false-negative count favors Logistic Regression on this test set.

13. ROC Curve Analysis

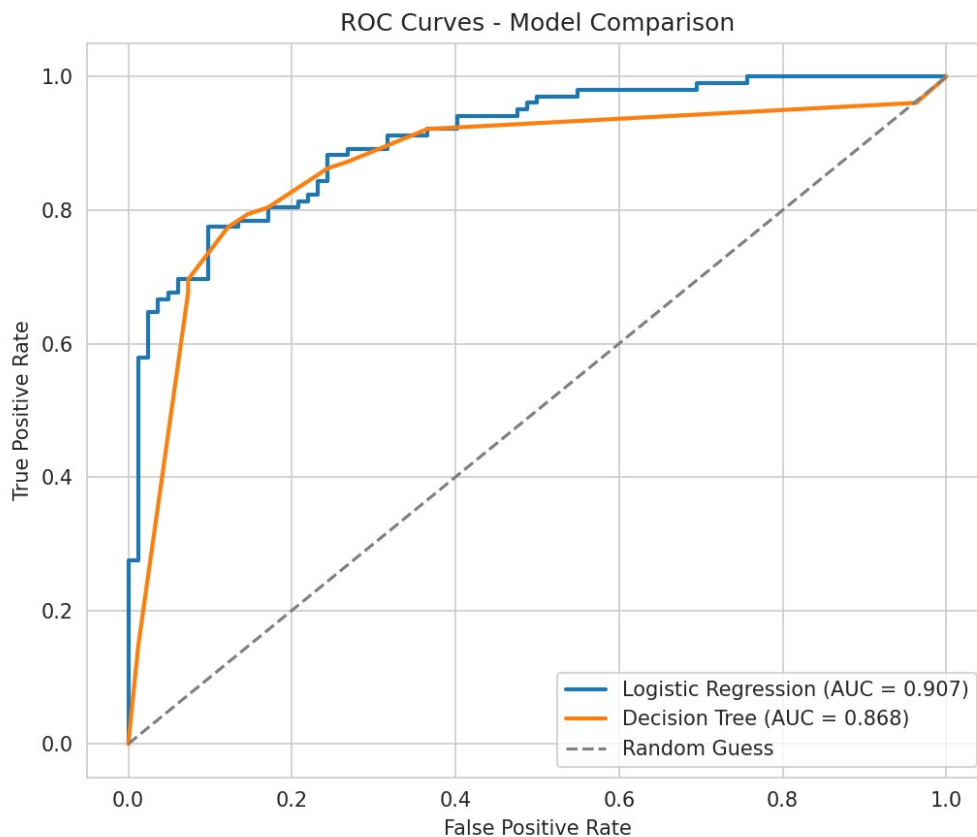


Figure 6. ROC curves for Logistic Regression and Decision Tree, with the random-guess diagonal shown for reference.

Both ROC curves are above the random-guess diagonal, indicating better-than-random discrimination on the held-out data. Logistic Regression has ROC AUC = 0.9073, while the Decision Tree has ROC AUC = 0.8684. This means Logistic Regression provided the better overall ordering of disease-present records above no-disease records across varying classification thresholds in the executed experiment.

14. Sample Prediction

The final saved model was Logistic Regression. The script demonstrated prediction on the first row of the held-out test subset, at original index 514. This example was not reported as a successful case; it is an observed misclassification from the executed output.

Sample field	Observed value
Held-out row index	514
Actual label	Disease (1)
Predicted label	No Disease (0)
Predicted probability of disease	0.3427110839545309

At the model's default decision rule, the predicted disease probability of 0.3427110839545309 resulted in a no-disease prediction even though the actual label was disease. This concrete example illustrates that the selected model is not error-free and that predictions close to a decision boundary require careful interpretation.

15. Limitations

- The raw dataset contains substantial missingness in clinically relevant columns, particularly ca (611 missing values) and thal (486 missing values). Mode imputation preserves rows but may weaken the true signal in these features.
- The project uses one stratified 80/20 split with random_state=42. It does not perform cross-validation, repeated validation, calibration analysis, or confidence-interval estimation.
- Only two classical models were evaluated: Logistic Regression and a depth-limited Decision Tree. No hyperparameter search or additional model families were executed.
- Categorical variables were label-encoded. This implementation choice creates numeric codes for categories and may not represent all categorical relationships optimally for Logistic Regression.
- The dataset combines multiple medical centers. Differences in collection practices may contribute to missingness and may affect generalizability.
- The binary target combines severity levels 1–4 into a single disease-present class, so the model does not predict disease severity.
- The project results are for a held-out dataset split and should not be interpreted as evidence of clinical readiness or external performance.

16. Ethical Considerations

- This model is an educational machine-learning project and must not be used as a standalone medical diagnostic tool.
- False negatives and false positives can both have meaningful consequences. Predictions should be reviewed by qualified healthcare professionals in any real-world setting.
- Clinical data may reflect differences in population coverage, collection sites, missingness patterns, and historical practice. Performance and fairness should be evaluated across relevant patient groups before deployment.
- Patient data requires strong privacy protection, controlled access, secure storage, and adherence to applicable healthcare and data-protection requirements.
- Model outputs should be communicated as probabilistic decision support, with clear explanation of uncertainty and limitations rather than as definitive disease labels.

17. Future Scope

- Evaluate more robust missing-data strategies, such as multiple imputation and missingness-indicator features, especially for ca and thal.
- Use cross-validation and a separate external validation dataset to obtain more reliable estimates of model generalization.
- Compare additional classifiers and conduct controlled hyperparameter tuning while keeping preprocessing inside an evaluation pipeline.
- Explore appropriate categorical encodings and model calibration, including threshold selection based on a clearly defined clinical use case.
- Assess subgroup performance and fairness once suitable demographic and outcome information is available.
- Develop interpretable model explanations and clinician-facing validation procedures before considering any decision-support application.

18. Conclusion

This internship capstone implemented a complete machine-learning pipeline for binary heart disease prediction using the supplied combined UCI dataset. The project cleaned 920 raw records, removed 2 duplicates, imputed missing values, encoded categorical variables, created a stratified train/test split, standardized features, and evaluated Logistic Regression and Decision Tree classifiers.

Both models achieved test accuracy of 0.8152. Logistic Regression was selected because it produced a higher ROC AUC of 0.9073 compared with 0.8684 for the Decision Tree, while also achieving higher recall of 0.8627. The results demonstrate a solid baseline model-comparison workflow. However, substantial missingness, limited validation, and the non-clinical nature of the exercise mean that the model should be treated as an educational prototype rather than a deployment-ready medical system.

19. References

- UCI Machine Learning Repository. Heart Disease dataset. Dataset used in this project as the supplied combined multi-center CSV file.

- Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- pandas development team. pandas: Python data analysis library.
- Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95.
- Waskom, M. L. (2021). seaborn: statistical data visualization. *Journal of Open Source Software*, 6(60), 3021.
- Project source artifacts reviewed for this report: `heart_disease_prediction.py`, `heart_disease_prediction.ipynb`, `results_summary.json`, `model_comparison.csv`, and the generated visualizations.